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| <div>UniSec</div> <div>Medium Voltage Switchgear</div> <div>PT MULTIMAS NABATI ASAHAN</div> <div>MVD-34 20kV FlourMill</div> <div>SCHEMATIC DIAGRAM</div> <div>SBC-W INCOMING</div> <div>H02</div> |                     |          |      |            |            |                                                     |          |                    |                        |      |                                                                             |   |              |                            |       | A | A |
|                                                                                                                                                                                                    |                     |          |      |            |            |                                                     |          |                    |                        |      |                                                                             |   |              |                            |       | B | B |
|                                                                                                                                                                                                    |                     |          |      |            |            |                                                     |          |                    |                        |      |                                                                             |   |              |                            |       | C | C |
|                                                                                                                                                                                                    |                     |          |      |            |            |                                                     |          |                    |                        |      |                                                                             |   |              |                            |       | D | D |
|                                                                                                                                                                                                    |                     |          |      |            |            |                                                     |          |                    |                        |      |                                                                             |   |              |                            |       | E | E |
| 1                                                                                                                                                                                                  |                     | 2        |      | 3          |            | 4                                                   |          | 5                  |                        | 6    |                                                                             | 7 |              | 8                          |       | F | F |
|                                                                                                                                                                                                    |                     |          |      | Date Prep. | 15.11.2018 | PROJECT TITLE<br>MV Switchgear for MNA Serang Plant |          |                    | ABB                    | EPMV | COVER SHEET<br>MVD-34 20kV FlourMill<br>SCHEMATIC DIAGRAM<br>SBC-W INCOMING |   | Scale        | NTS                        | = H02 |   |   |
|                                                                                                                                                                                                    |                     |          |      | Prepared   | YR         |                                                     |          |                    |                        |      |                                                                             |   | Customer     | PT WILMAR NABATI INDONESIA | & 304 |   |   |
| 00                                                                                                                                                                                                 | Issued for Approval | 22.02.19 | YR   | Checked    | AW         |                                                     |          |                    | PT. ABB Sakti Industri |      |                                                                             |   | Drawing Nbr. | 18.042-WNI-304.02          | Sh.   | 1 |   |
| Ind.                                                                                                                                                                                               | Revision            | Date     | Name | Approved   | AI         | Based on:                                           | BE318043 | Customer Doc. Nbr. |                        |      |                                                                             |   | Cont.        | 28                         |       |   |   |

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| A                   | SHEET INDEX         |                     |                                     |                                     |                                     |            |                                    |          | A                  |                   |                                                                                |          |                                  |       |
|                     | PAGE                |                     |                                     | PAGE NAME                           |                                     |            |                                    | REVISION |                    |                   |                                                                                |          |                                  |       |
|                     | ==MVD-34=H02&304/1  |                     |                                     | COVER SHEET                         |                                     |            |                                    | 00       |                    |                   |                                                                                |          |                                  |       |
|                     | ==MVD-34=H02&304/2  |                     |                                     | Index of Sheet                      |                                     |            |                                    | 00       |                    |                   |                                                                                |          |                                  |       |
|                     | ==MVD-34=H02&304/3  |                     |                                     | REFERENCE DESIGNATIONS              |                                     |            |                                    | 00       |                    |                   |                                                                                |          |                                  |       |
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| E                   |                     |                     |                                     |                                     |                                     |            |                                    |          | E                  |                   |                                                                                |          |                                  |       |
|                     |                     |                     |                                     |                                     | Date Prep.                          | 15.11.2018 | PROJECT TITLE                      |          | ABB                | EPMV              | Index of Sheet<br>MVD-34 20kV FlourMill<br>SCHEMATIC DIAGRAM<br>SBC-W INCOMING | Scale    | NTS                              | = H02 |
|                     |                     |                     |                                     |                                     | Prepared                            | YR         | MV Switchgear for MNA Serang Plant |          |                    |                   |                                                                                | Customer | PT WILMAR NABATI INDONESIA & 304 |       |
|                     | 00                  | Issued for Approval | 22.02.19                            | YR                                  | Checked                             | AW         |                                    |          | Drawing Nbr.       | 18.042-WNI-304.02 |                                                                                | Sh.      | 2                                |       |
|                     | Ind.                | Revision            | Date                                | Name                                | Approved                            | AI         | Based on: BE318043                 |          | Customer Doc. Nbr. |                   |                                                                                | Cont.    | 28                               |       |
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|---|-------|----------------------------------------------------------------------------------------------------------------|--|--|--|--|--|--------|--------------------------------------------------------------------------------------------------------------|--|--|--|--|--|
| A | -MAE  | MOTOR FOR ELECTRICAL OPERATION OF EARTHING SWITCH -QCE                                                         |  |  |  |  |  | -QBH   | MANUAL CIRCUIT-BREAKERS                                                                                      |  |  |  |  |  |
|   | -MAE1 | MOTOR FOR ELECTRICAL OPERATION OF EARTHING SWITCH -QCE1 OR -QCE2                                               |  |  |  |  |  | -QBL   | LINKS                                                                                                        |  |  |  |  |  |
|   | -MAE8 | MOTOR FOR ELECTRICAL OPERATION OF EARTHING SWITCH -QCE SUPPLIED TOGETHER WITH SWITCH-DISCONNECTOR -QBS         |  |  |  |  |  | -QBM   | MINIATURE SWITCH-DISCONNECTORS                                                                               |  |  |  |  |  |
|   | -MAS  | MOTOR FOR CIRCUIT-BREAKER SPRINGS CHARGING                                                                     |  |  |  |  |  | -QBS   | SWITCH-DISCONNECTORS                                                                                         |  |  |  |  |  |
|   | -MAT  | MOTOR FOR ELECTRICAL OPERATION OF TRUCK RACKING-IN/OUT                                                         |  |  |  |  |  | -QBT   | TRUCK                                                                                                        |  |  |  |  |  |
|   | -MBC  | CLOSING RELEASE OF CIRCUIT-BREAKER                                                                             |  |  |  |  |  | -QCA   | EARTHING SWITCH FOR ELIMINATION OF INTERNAL ARCING                                                           |  |  |  |  |  |
|   | -MBC4 | CLOSING RELEASE OF SWITCH-DISCONNECTOR -QBS                                                                    |  |  |  |  |  | -QCE   | EARTHING SWITCH                                                                                              |  |  |  |  |  |
|   | -MBO1 | FIRST OPENING RELEASE OF CIRCUIT-BREAKER                                                                       |  |  |  |  |  | -QCE1  | EARTHING SWITCH OF BUSBARS (SINGLE BUSBAR SYSTEM) OR OF FRONT BUSBARS (DOUBLE BUSBAR SYSTEM)                 |  |  |  |  |  |
|   | -MBO2 | SECOND OPENING RELEASE OF CIRCUIT-BREAKER                                                                      |  |  |  |  |  | -QCE2  | EARTHING SWITCH OF REAR BUSBARS (DOUBLE BUSBAR SYSTEM)                                                       |  |  |  |  |  |
|   | -MBO3 | OPENING SOLENOID FOR OVERCURRENT RELEASE OF CIRCUIT-BREAKER OR INDIRECT OVERCURRENT RELEASE ON CIRCUIT-BREAKER |  |  |  |  |  | -QQ    | CLUTCHES                                                                                                     |  |  |  |  |  |
| B | -MBO4 | OPENING RELEASE OF SWITCH-DISCONNECTOR -QBS                                                                    |  |  |  |  |  | -RAA   | ANTIFERRORESONANCE RESISTOR                                                                                  |  |  |  |  |  |
|   | -MBU  | UNDERVOLTAGE RELEASE OF CIRCUIT-BREAKER                                                                        |  |  |  |  |  | -RAD   | DIODES                                                                                                       |  |  |  |  |  |
|   | -PFB  | BLUE SIGNAL LAMPS                                                                                              |  |  |  |  |  | -RAI   | INDUCTORS                                                                                                    |  |  |  |  |  |
|   | -PFF  | FLAG RELAYS                                                                                                    |  |  |  |  |  | -RAR   | RESISTORS                                                                                                    |  |  |  |  |  |
|   | -PFG  | GREEN SIGNAL LAMPS                                                                                             |  |  |  |  |  | -RF    | FILTERS                                                                                                      |  |  |  |  |  |
| C | -PFR  | RED SIGNAL LAMPS                                                                                               |  |  |  |  |  | -RLE1  | ELECTROMECHANICAL LOCK PREVENTING CIRCUIT-BREAKER CLOSING                                                    |  |  |  |  |  |
|   | -PFS  | SHORT CIRCUIT INDICATORS                                                                                       |  |  |  |  |  | -RLE2  | ELECTROMECHANICAL LOCK PREVENTING TRUCK RACKING-IN/OUT                                                       |  |  |  |  |  |
|   | -PFV  | VOLTAGE INDICATORS                                                                                             |  |  |  |  |  | -RLE3  | ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR CLOSING OPERATION OF EARTHING SWITCH -QCE           |  |  |  |  |  |
|   | -PFV1 | VOLTAGE INDICATORS ON FEEDER SIDE                                                                              |  |  |  |  |  | -RLE4  | ELECTROMECHANICAL LOCK PREVENTING THE DOOR OPENING OPERATION                                                 |  |  |  |  |  |
|   | -PFV2 | VOLTAGE INDICATORS ON BUSBAR SIDE                                                                              |  |  |  |  |  | -RLE5  | ELECTROMECHANICAL LOCK PREVENTING OPERATION OF DISCONNECTOR -QBD                                             |  |  |  |  |  |
|   | -PFW  | WHITE SIGNAL LAMPS                                                                                             |  |  |  |  |  | -RLE6  | ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR CLOSING OPERATION OF DISCONNECTOR -QBD1             |  |  |  |  |  |
|   | -PFX  | CROSS INDICATORS, ELECTROMECHANICAL INDICATORS                                                                 |  |  |  |  |  | -RLE7  | ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR CLOSING OPERATION OF DISCONNECTOR -QBD2             |  |  |  |  |  |
|   | -PFY  | YELLOW SIGNAL LAMPS                                                                                            |  |  |  |  |  | -RLE8  | ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR CLOSING OPERATION OF EARTHING SWITCH -QCE1 OR -QCE2 |  |  |  |  |  |
|   | -PGA  | AMMETERS                                                                                                       |  |  |  |  |  | -RLE9  | ELECTROMECHANICAL LOCK PREVENTING OPERATION OF SWITCH-DISCONNECTOR -QBS                                      |  |  |  |  |  |
|   | -PGC  | COUNTERS                                                                                                       |  |  |  |  |  | -RLE11 | ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR OPERATION OF DISCONNECTOR -QBD4                     |  |  |  |  |  |
| D | -PGF  | FREQUENCYMETERS                                                                                                |  |  |  |  |  | -RLE12 | ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR OPERATION OF DISCONNECTOR -QBD5                     |  |  |  |  |  |
|   | -PGH  | HOURMETERS                                                                                                     |  |  |  |  |  | -SFA   | AMMETRIC SWITCHES                                                                                            |  |  |  |  |  |
|   | -PGI  | PROTECTION AND CONTROL UNIT: HUMAN MACHINE INTERFACE                                                           |  |  |  |  |  | -SFC   | CONTROL SWITCHES, CLOSING PUSH-BUTTONS                                                                       |  |  |  |  |  |
|   | -PGI1 | HUMAN MACHINE INTERFACE OF ELECTRONIC CONTROLLED CIRCUIT-BREAKER (ON CIRCUIT-BREAKER)                          |  |  |  |  |  | -SFL   | LOCKING CONTACTS                                                                                             |  |  |  |  |  |
|   | -PGI2 | HUMAN MACHINE INTERFACE FOR ELECTRONIC CONTROLLED CIRCUIT-BREAKER (ON SWITCHGEAR)                              |  |  |  |  |  | -SFO   | OPENING PUSH-BUTTONS                                                                                         |  |  |  |  |  |
|   | -PGJ  | ACTIVE ENERGY METERS                                                                                           |  |  |  |  |  | -SFR   | RESET PUSH-BUTTONS                                                                                           |  |  |  |  |  |
|   | -PGK  | REACTIVE ENERGY METERS                                                                                         |  |  |  |  |  | -SFS   | SELECTOR SWITCHES                                                                                            |  |  |  |  |  |
|   | -PGM  | MULTIFUNCTION INDICATORS                                                                                       |  |  |  |  |  | -SFT   | TEST PUSH-BUTTONS                                                                                            |  |  |  |  |  |
|   | -PGP  | POWER-FACTOR METERS                                                                                            |  |  |  |  |  | -SFU   | UNLOCKING PUSH-BUTTONS                                                                                       |  |  |  |  |  |
|   | -PGQ  | VARMETERS                                                                                                      |  |  |  |  |  | -SFV   | VOLTMETRIC SWITCHES                                                                                          |  |  |  |  |  |
| E | -PGS  | SYNCHRONOSCOPES                                                                                                |  |  |  |  |  | -TA    | POWER TRANSFORMERS                                                                                           |  |  |  |  |  |
|   | -PGV  | VOLTMETERS                                                                                                     |  |  |  |  |  | -TB    | CONVERTER                                                                                                    |  |  |  |  |  |
|   | -PGW  | WATTMETERS                                                                                                     |  |  |  |  |  | -TB1   | RECTIFIER FOR FIRST OPENING RELEASE                                                                          |  |  |  |  |  |
|   | -PJ   | ACOUSTICAL SIGNAL DEVICES (BELLS, SIRENS)                                                                      |  |  |  |  |  | -TB2   | RECTIFIER FOR SECOND OPENING RELEASE                                                                         |  |  |  |  |  |
|   | -QAB  | CIRCUIT-BREAKERS                                                                                               |  |  |  |  |  | -TB3   | RECTIFIER FOR CLOSING RELEASE                                                                                |  |  |  |  |  |
|   | -QAC  | CONTACTORS (FOR POWER)                                                                                         |  |  |  |  |  | -TB4   | RECTIFIER FOR ELECTROMECHANICAL LOCK PREVENTING CIRCUIT-BREAKER CLOSING                                      |  |  |  |  |  |
|   | -QBD  | DISCONNECTORS                                                                                                  |  |  |  |  |  | -TB5   | RECTIFIER FOR ELECTROMECHANICAL LOCK PREVENTING TRUCK RACKING-IN/OUT                                         |  |  |  |  |  |
|   | -QBD1 | FRONT BUSBAR DISCONNECTORS (DOUBLE BUSBAR SYSTEM)                                                              |  |  |  |  |  | -TB6   | RECTIFIER FOR UNDERVOLTAGE RELEASE                                                                           |  |  |  |  |  |
|   | -QBD2 | REAR BUSBAR DISCONNECTORS (DOUBLE BUSBAR SYSTEM)                                                               |  |  |  |  |  | -TB7   | ERECTIFIER FOR INDIRECT OVERCURRENT RELEASE ON CIRCUIT-BREAKER                                               |  |  |  |  |  |
|   | -QBD3 | FEEDER DISCONNECTORS (DOUBLE BUSBAR SYSTEM)                                                                    |  |  |  |  |  |        |                                                                                                              |  |  |  |  |  |
| F | -QBD4 | TWO-WAY DISCONNECTOR                                                                                           |  |  |  |  |  |        |                                                                                                              |  |  |  |  |  |
|   | -QBD5 | TWO-WAY DISCONNECTOR ON BUSBAR RISER                                                                           |  |  |  |  |  |        |                                                                                                              |  |  |  |  |  |
|   |       |                                                                                                                |  |  |  |  |  |        |                                                                                                              |  |  |  |  |  |
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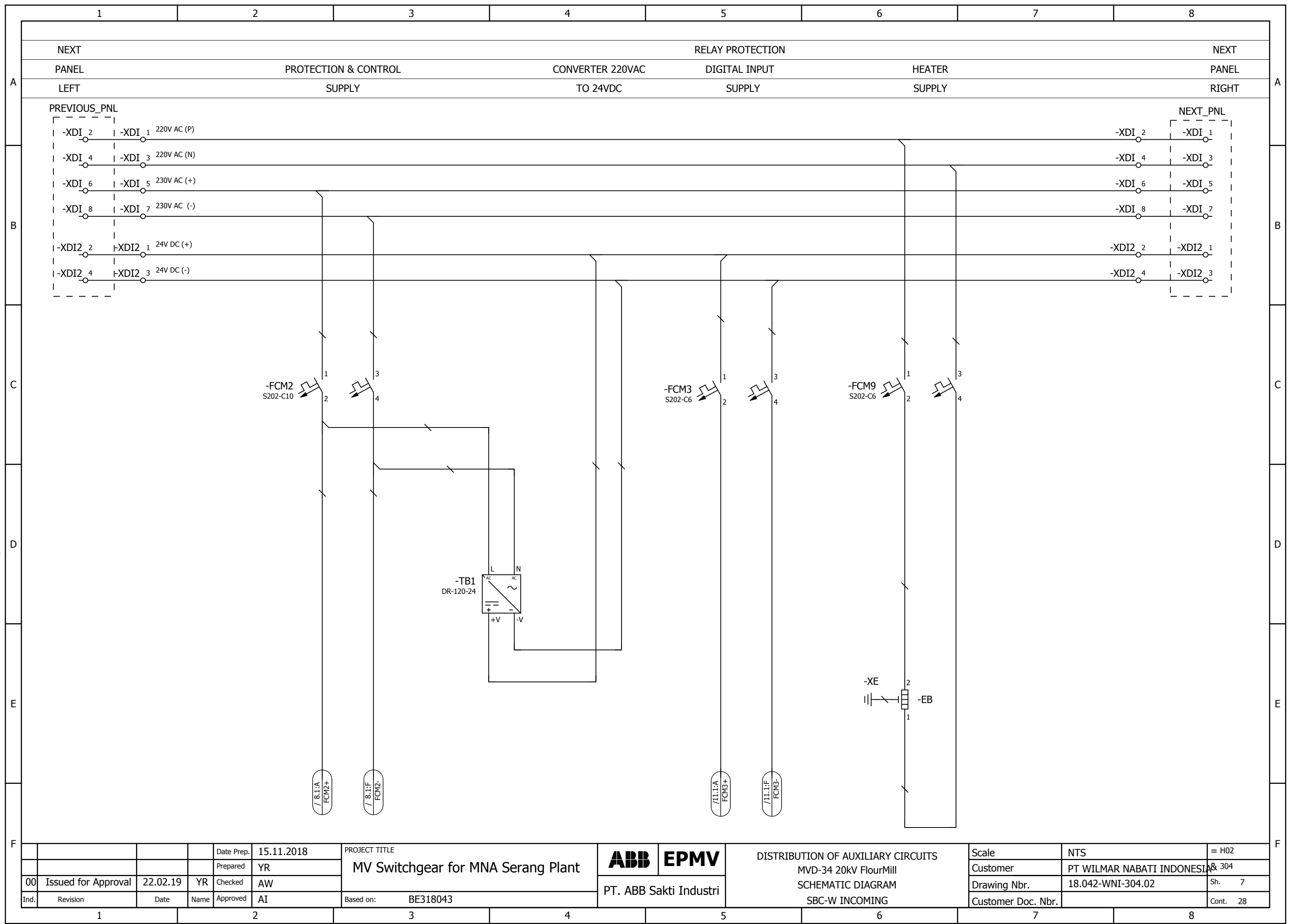
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| A               | -TFA                                                                                               | ACTIVE POWER TRANSDUCERS                                           | -XDI1                                               | TERMINAL BLOCK FOR INTERCONNECTIONS OF CONTROL CIRCUITS            | A                                                                  |                                                                                               |                                                                                            |                                                                   |                                                                                    |                                                         |                                                                     |                                                                     |                                                                        |
|                 | -TFC                                                                                               | CURRENT TRANSDUCERS                                                |                                                     | -XDI2                                                              |                                                                    | TERMINAL BLOCK FOR INTERCONNECTIONS OF CIRCUITS OF MOTOR FOR CIRCUIT-BREAKER SPRINGS CHARGING |                                                                                            |                                                                   |                                                                                    |                                                         |                                                                     |                                                                     |                                                                        |
|                 | -TFF                                                                                               | FREQUENCY TRANSDUCERS                                              |                                                     |                                                                    |                                                                    | -XDI3                                                                                         | TERMINAL BLOCK FOR INTERCONNECTIONS OF CIRCUITS OF MOTOR FOR SWITCH-DISCONNECTOR OPERATION |                                                                   |                                                                                    |                                                         |                                                                     |                                                                     |                                                                        |
|                 | -TFJ                                                                                               | ACTIVE ENERGY TRANSDUCERS                                          |                                                     |                                                                    |                                                                    |                                                                                               | -XDI4                                                                                      | TERMINAL BLOCK FOR INTERCONNECTIONS OF A.C. AUXILIARY CIRCUITS    |                                                                                    |                                                         |                                                                     |                                                                     |                                                                        |
|                 | -TFK                                                                                               | REACTIVE ENERGY TRANSDUCERS                                        |                                                     |                                                                    |                                                                    |                                                                                               |                                                                                            | -XDI6                                                             | TERMINAL BLOCK FOR INTERCONNECTION (CONNECTION BETWEEN PANELS) OF VOLTAGE CIRCUITS |                                                         |                                                                     |                                                                     |                                                                        |
|                 | -TFM                                                                                               | MULTIFUNCTION TRANSDUCERS                                          |                                                     |                                                                    |                                                                    |                                                                                               |                                                                                            |                                                                   |                                                                                    |                                                         |                                                                     |                                                                     |                                                                        |
|                 | -TFP                                                                                               | POWER-FACTOR TRANSDUCERS                                           |                                                     |                                                                    |                                                                    |                                                                                               |                                                                                            |                                                                   |                                                                                    |                                                         |                                                                     |                                                                     |                                                                        |
|                 | -TFQ                                                                                               | REACTIVE POWER TRANSDUCERS                                         |                                                     |                                                                    |                                                                    |                                                                                               |                                                                                            |                                                                   |                                                                                    |                                                         |                                                                     |                                                                     |                                                                        |
|                 | -TFS                                                                                               | SIGNAL CONVERTERS                                                  |                                                     |                                                                    |                                                                    |                                                                                               |                                                                                            |                                                                   |                                                                                    |                                                         |                                                                     |                                                                     |                                                                        |
|                 | -TFV                                                                                               | VOLTAGE TRANSDUCERS                                                |                                                     |                                                                    |                                                                    |                                                                                               |                                                                                            |                                                                   |                                                                                    |                                                         |                                                                     |                                                                     |                                                                        |
| B               | -WA                                                                                                | BUSBARS                                                            | -XDI7                                               |                                                                    | TERMINAL BLOCK FOR INTERCONNECTION OF MOD-BUS                      |                                                                                               |                                                                                            |                                                                   | B                                                                                  |                                                         |                                                                     |                                                                     |                                                                        |
|                 | -WBC                                                                                               | POWER CABLES                                                       |                                                     | -XDI8                                                              | TERMINAL BLOCK FOR INTERCONNECTIONS OF CURRENT CIRCUITS            |                                                                                               |                                                                                            |                                                                   |                                                                                    |                                                         |                                                                     |                                                                     |                                                                        |
|                 | -WE                                                                                                | EARTHING CABLES FOR L.V. INSTRUMENT COMPARTMENT.                   |                                                     |                                                                    | -XDI9                                                              | TERMINAL BLOCK FOR SPECIAL USE                                                                |                                                                                            |                                                                   |                                                                                    |                                                         |                                                                     |                                                                     |                                                                        |
|                 | -WGC                                                                                               | CONTROL CABLES                                                     |                                                     |                                                                    |                                                                    | -XDJ                                                                                          | TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF DISCONNECTORS -QBD                                |                                                                   |                                                                                    |                                                         |                                                                     |                                                                     |                                                                        |
|                 | -WGS                                                                                               | COMMUNICATION CABLES (SHIELDED TWISTED PAIRS)                      |                                                     |                                                                    |                                                                    |                                                                                               | -XDJ1                                                                                      | TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF SWITCH-DISCONNECTOR -QBS |                                                                                    |                                                         |                                                                     |                                                                     |                                                                        |
|                 | -WH                                                                                                | OPTICAL FIBRES                                                     |                                                     |                                                                    |                                                                    |                                                                                               |                                                                                            | -XDM                                                              |                                                                                    | SEALABLE TERMINAL BLOCK FOR MEASUREMENT                 |                                                                     |                                                                     |                                                                        |
|                 | -XDA                                                                                               | TERMINAL BLOCK FOR CIRCUITS OF CURRENT TRANSFORMERS                |                                                     |                                                                    |                                                                    |                                                                                               |                                                                                            |                                                                   |                                                                                    | -XDS                                                    | CURRENT SOCKETS                                                     |                                                                     |                                                                        |
|                 | -XDA1                                                                                              | CONNECTOR FOR CIRCUITS OF CURRENT TRANSFORMERS LOCATED ON PHASE L1 |                                                     |                                                                    |                                                                    |                                                                                               |                                                                                            |                                                                   |                                                                                    |                                                         | -XDT                                                                | TERMINAL BLOCK FOR POSITION CONTACTS OF TRUCK AND OF CONNECTOR -XDB |                                                                        |
|                 | -XDA2                                                                                              | CONNECTOR FOR CIRCUITS OF CURRENT TRANSFORMERS LOCATED ON PHASE L2 |                                                     |                                                                    |                                                                    |                                                                                               |                                                                                            |                                                                   |                                                                                    |                                                         |                                                                     | -XDU                                                                | CONNECTOR FOR ISOLATION OF CIRCUITS OF LOWER VOLTAGE TRANSFORMER TRUCK |
|                 | -XDA3                                                                                              | CONNECTOR FOR CIRCUITS OF CURRENT TRANSFORMERS LOCATED ON PHASE L3 |                                                     |                                                                    |                                                                    |                                                                                               |                                                                                            |                                                                   |                                                                                    |                                                         |                                                                     |                                                                     | -XDU1                                                                  |
| -XDA5           | CONNECTOR FOR CIRCUITS OF NEUTRAL (RESIDUAL) CURRENT TRANSFORMER                                   | -XDV                                                               | TERMINAL BLOCK FOR CIRCUITS OF VOLTAGE TRANSFORMERS |                                                                    |                                                                    |                                                                                               |                                                                                            |                                                                   |                                                                                    |                                                         |                                                                     |                                                                     |                                                                        |
| -XDB            | CONNECTOR FOR ISOLATION OF CIRCUIT-BREAKER (OR UPPER VOLTAGE TRANSFORMER TRUCK) AUXILIARY CIRCUITS |                                                                    | -XDV1                                               | CONNECTOR FOR CIRCUITS OF VOLTAGE TRANSFORMERS LOCATED ON PHASE L1 |                                                                    |                                                                                               |                                                                                            |                                                                   |                                                                                    |                                                         |                                                                     |                                                                     |                                                                        |
| -XDB1           | CONNECTOR FOR AUXILIARY CIRCUITS OF CIRCUIT-BREAKER (OR UPPER VOLTAGE TRANSFORMER TRUCK)           |                                                                    |                                                     | -XDV2                                                              | CONNECTOR FOR CIRCUITS OF VOLTAGE TRANSFORMERS LOCATED ON PHASE L2 |                                                                                               |                                                                                            |                                                                   |                                                                                    |                                                         |                                                                     |                                                                     |                                                                        |
| -XDB2           | CONNECTOR FOR ADDITIONAL AUXILIARY CIRCUITS OF CIRCUIT-BREAKER                                     |                                                                    |                                                     |                                                                    | -XDV3                                                              | CONNECTOR FOR CIRCUITS OF VOLTAGE TRANSFORMERS LOCATED ON PHASE L3                            |                                                                                            |                                                                   |                                                                                    |                                                         |                                                                     |                                                                     |                                                                        |
| -XDB10...-XDB89 | CONNECTOR FOR CIRCUIT-BREAKER INTERNAL CIRCUITS                                                    |                                                                    |                                                     |                                                                    |                                                                    | -XDV4                                                                                         | CONNECTOR FOR CIRCUITS OF ANTIFERRORESONANCE RESISTOR                                      |                                                                   |                                                                                    |                                                         |                                                                     |                                                                     |                                                                        |
| -XDB7           | CONNECTOR FOR TRUCK POSITION CONTACTS                                                              |                                                                    |                                                     |                                                                    |                                                                    |                                                                                               | -XDX                                                                                       | SUPPORT TERMINAL BLOCK                                            |                                                                                    |                                                         |                                                                     |                                                                     |                                                                        |
| -XDB91          | CONNECTOR FOR OPENING RELEASE CIRCUITS OF SWITCH-DISCONNECTOR                                      |                                                                    |                                                     |                                                                    |                                                                    |                                                                                               |                                                                                            | -XE1                                                              | EARTHING TERMINAL BLOCKS FOR LV COMPARTMENT, LEFT SIDE                             |                                                         |                                                                     |                                                                     |                                                                        |
| -XDB92          | CONNECTOR FOR CLOSING RELEASE CIRCUITS OF SWITCH-DISCONNECTOR                                      |                                                                    |                                                     |                                                                    |                                                                    |                                                                                               |                                                                                            |                                                                   | -XE2                                                                               | EARTHING TERMINAL BLOCKS FOR LV COMPARTMENT, RIGHT SIDE |                                                                     |                                                                     |                                                                        |
| -XDB93          | CONNECTOR FOR POSITION SWITCHES OF SWITCH-DISCONNECTOR                                             |                                                                    |                                                     |                                                                    |                                                                    |                                                                                               |                                                                                            |                                                                   |                                                                                    | -XE5                                                    | EARTHING TERMINAL BLOCKS FOR MV COMPARTMENT, VOLTAGE INDICATOR SIDE |                                                                     |                                                                        |
| -XDB95          | CONNECTOR FOR CIRCUITS OF EARTHING SWITCH LOCKING MAGNET                                           |                                                                    |                                                     |                                                                    |                                                                    |                                                                                               |                                                                                            |                                                                   |                                                                                    |                                                         | -XE6                                                                | EARTHING TERMINAL BLOCKS FOR MV COMPARTMENT, REAR BOTTOM SIDE       |                                                                        |
| -XDC            | CUSTOMER TERMINAL BLOCK                                                                            | -XE7                                                               |                                                     |                                                                    |                                                                    |                                                                                               |                                                                                            |                                                                   |                                                                                    |                                                         |                                                                     | EARTHING TERMINAL BLOCKS FOR MV COMPARTMENT, LEFT SIDE              |                                                                        |
| -XDC1           | CUSTOMER TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF CIRCUIT BREAKER                                  |                                                                    | -XE8                                                |                                                                    |                                                                    |                                                                                               |                                                                                            |                                                                   |                                                                                    |                                                         |                                                                     | EARTHING TERMINAL BLOCKS FOR MV COMPARTMENT, FRONT BOTTOM SIDE      |                                                                        |
| -XDC2           | CUSTOMER TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF SWITCH-DISCONNECTOR                              |                                                                    |                                                     | -XE9                                                               |                                                                    |                                                                                               |                                                                                            |                                                                   |                                                                                    |                                                         |                                                                     | EARTHING TERMINAL BLOCK FOR TRANSFORMERS IN BUSBAR COMPARTMENT      |                                                                        |
| -XDC3           | CUSTOMER TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF PROTECTION RELAY AND MULTIFUNCTION UNIT          |                                                                    |                                                     |                                                                    |                                                                    |                                                                                               |                                                                                            |                                                                   |                                                                                    |                                                         |                                                                     |                                                                     |                                                                        |
| -XDC4           | CUSTOMER TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF VOLTAGE TRANSFORMERS                             |                                                                    |                                                     |                                                                    |                                                                    |                                                                                               |                                                                                            |                                                                   |                                                                                    |                                                         |                                                                     |                                                                     |                                                                        |
| -XDC5           | CUSTOMER TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF MINIATURE CIRCUIT BREAKER                        |                                                                    |                                                     |                                                                    |                                                                    |                                                                                               |                                                                                            |                                                                   |                                                                                    |                                                         |                                                                     |                                                                     |                                                                        |
| -XDE            | TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF EARTHING SWITCH -QCE                                      |                                                                    |                                                     |                                                                    |                                                                    |                                                                                               |                                                                                            |                                                                   |                                                                                    |                                                         |                                                                     |                                                                     |                                                                        |
| -XDE1           | TERMINAL BLOCK FOR INTERNAL CIRCUITS OF EARTHING SWITCH -QCE                                       |                                                                    |                                                     |                                                                    |                                                                    |                                                                                               |                                                                                            |                                                                   |                                                                                    |                                                         |                                                                     |                                                                     |                                                                        |
| -XDE2           | TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF EARTHING SWITCH -QCE1 OR -QCE2                            |                                                                    |                                                     |                                                                    |                                                                    |                                                                                               |                                                                                            |                                                                   |                                                                                    |                                                         |                                                                     |                                                                     |                                                                        |
| -XDE3           | TERMINAL BLOCK FOR INTERNAL CIRCUITS OF EARTHING SWITCH -QCE1 OR -QCE2                             |                                                                    |                                                     |                                                                    |                                                                    |                                                                                               |                                                                                            |                                                                   |                                                                                    |                                                         |                                                                     |                                                                     |                                                                        |
| -XDF            | TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF FORCED COOLING VENTILATORS                                |                                                                    |                                                     |                                                                    |                                                                    |                                                                                               |                                                                                            |                                                                   |                                                                                    |                                                         |                                                                     |                                                                     |                                                                        |
| -XDF1           | CONNECTOR FOR ISOLATION OF CIRCUITS OF FRONT VENTILATOR                                            |                                                                    |                                                     |                                                                    |                                                                    |                                                                                               |                                                                                            |                                                                   |                                                                                    |                                                         |                                                                     |                                                                     |                                                                        |
| -XDF2           | CONNECTOR FOR REAR VENTILATOR CIRCUITS                                                             |                                                                    |                                                     |                                                                    |                                                                    |                                                                                               |                                                                                            |                                                                   |                                                                                    |                                                         |                                                                     |                                                                     |                                                                        |
| -XDF3           | CONNECTOR FOR FRONT VENTILATOR CIRCUITS                                                            |                                                                    |                                                     |                                                                    |                                                                    |                                                                                               |                                                                                            |                                                                   |                                                                                    |                                                         |                                                                     |                                                                     |                                                                        |
| -XDH            | TERMINAL BLOCK FOR VARIOUS CIRCUITS (HEATERS, POSITION SWITCHES DETECTING INTERNAL ARCING, ETC.)   |                                                                    |                                                     |                                                                    |                                                                    |                                                                                               |                                                                                            |                                                                   |                                                                                    |                                                         |                                                                     |                                                                     |                                                                        |
| -XDI            | TERMINAL BLOCK FOR INTERCONNECTION (CONNECTION BETWEEN PANELS)                                     |                                                                    |                                                     |                                                                    |                                                                    |                                                                                               |                                                                                            |                                                                   |                                                                                    |                                                         |                                                                     |                                                                     |                                                                        |

|      |                     |          |      |            |            |                                                     |          |                        |      |                                                                                        |  |              |                            |       |
|------|---------------------|----------|------|------------|------------|-----------------------------------------------------|----------|------------------------|------|----------------------------------------------------------------------------------------|--|--------------|----------------------------|-------|
|      |                     |          |      | Date Prep. | 15.11.2018 | PROJECT TITLE<br>MV Switchgear for MNA Serang Plant |          | ABB                    | EPMV | REFERENCE DESIGNATIONS<br>MVD-34 20kV FlourMill<br>SCHEMATIC DIAGRAM<br>SBC-W INCOMING |  | Scale        | NTS                        | = H02 |
|      |                     |          |      | Prepared   | YR         |                                                     |          |                        |      |                                                                                        |  | Customer     | PT WILMAR NABATI INDONESIA | & 304 |
| 00   | Issued for Approval | 22.02.19 | YR   | Checked    | AW         |                                                     |          |                        |      |                                                                                        |  | Drawing Nbr. | 18.042-WNI-304.02          | Sh.   |
| Ind. | Revision            | Date     | Name | Approved   | AI         | Based on:                                           | BE318043 | PT. ABB Sakti Industri |      | Customer Doc. Nbr.                                                                     |  |              | Cont.                      | 28    |

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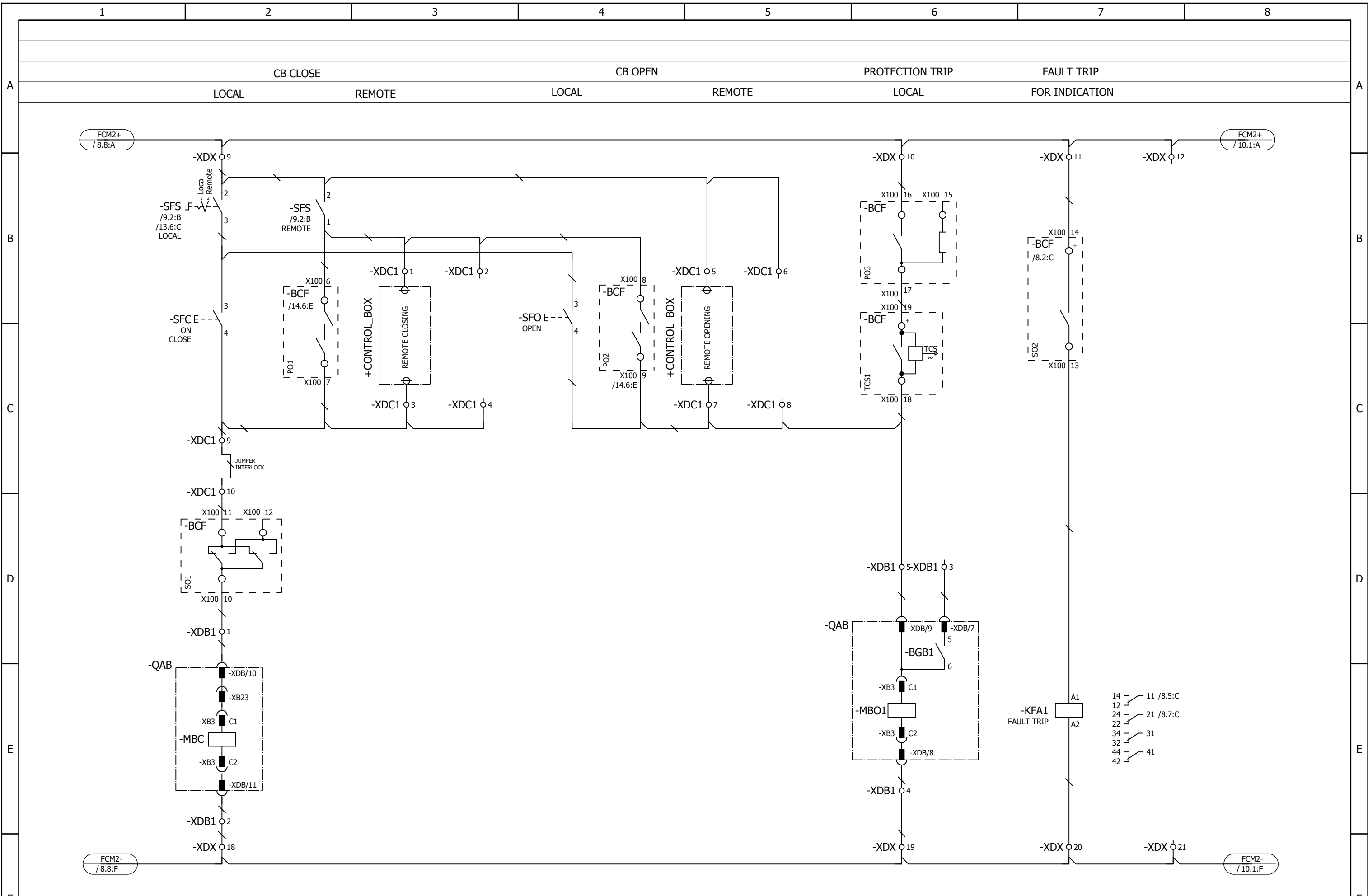








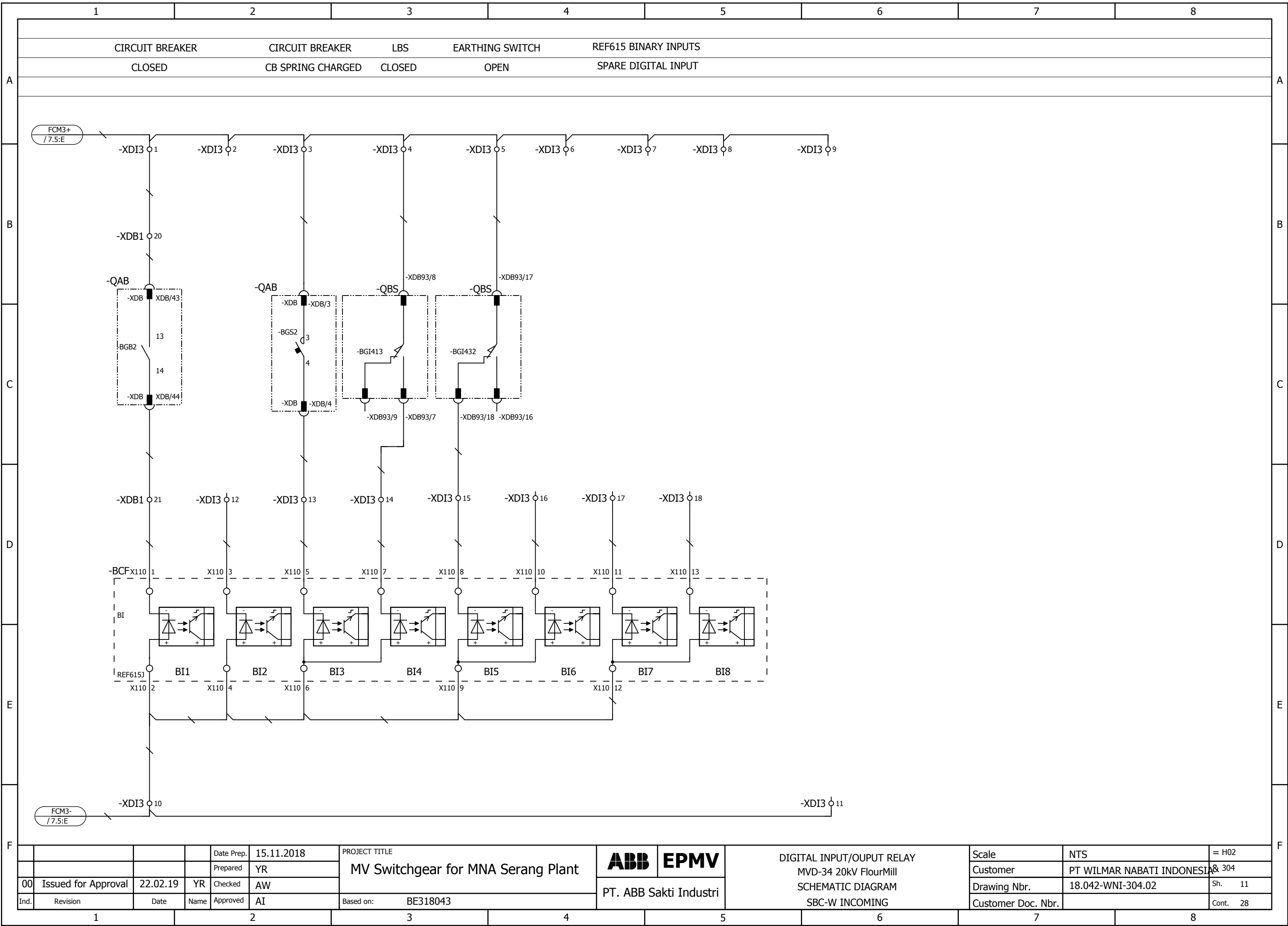
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|                        |          |          |      | Date Prep. | 15.11.2018 | PROJECT TITLE<br>MV Switchgear for MNA Serang Plant |          | ABB EPMV<br>PT. ABB Sakti Industri |  | CIRCUIT BREAKER CONTROL<br>MVD-34 20kV FlourMill<br>SCHEMATIC DIAGRAM<br>SBC-W INCOMING |  | Scale              | NTS                        | = H02    |
|                        |          |          |      | Prepared   | YR         |                                                     |          |                                    |  |                                                                                         |  | Customer           | PT WILMAR NABATI INDONESIA | & 304    |
| 00 Issued for Approval |          | 22.02.19 | YR   | Checked    | AW         |                                                     |          |                                    |  |                                                                                         |  | Drawing Nbr.       | 18.042-WNI-304.02          | Sh. 9    |
| Ind.                   | Revision | Date     | Name | Approved   | AI         | Based on:                                           | BE318043 |                                    |  |                                                                                         |  | Customer Doc. Nbr. |                            | Cont. 28 |

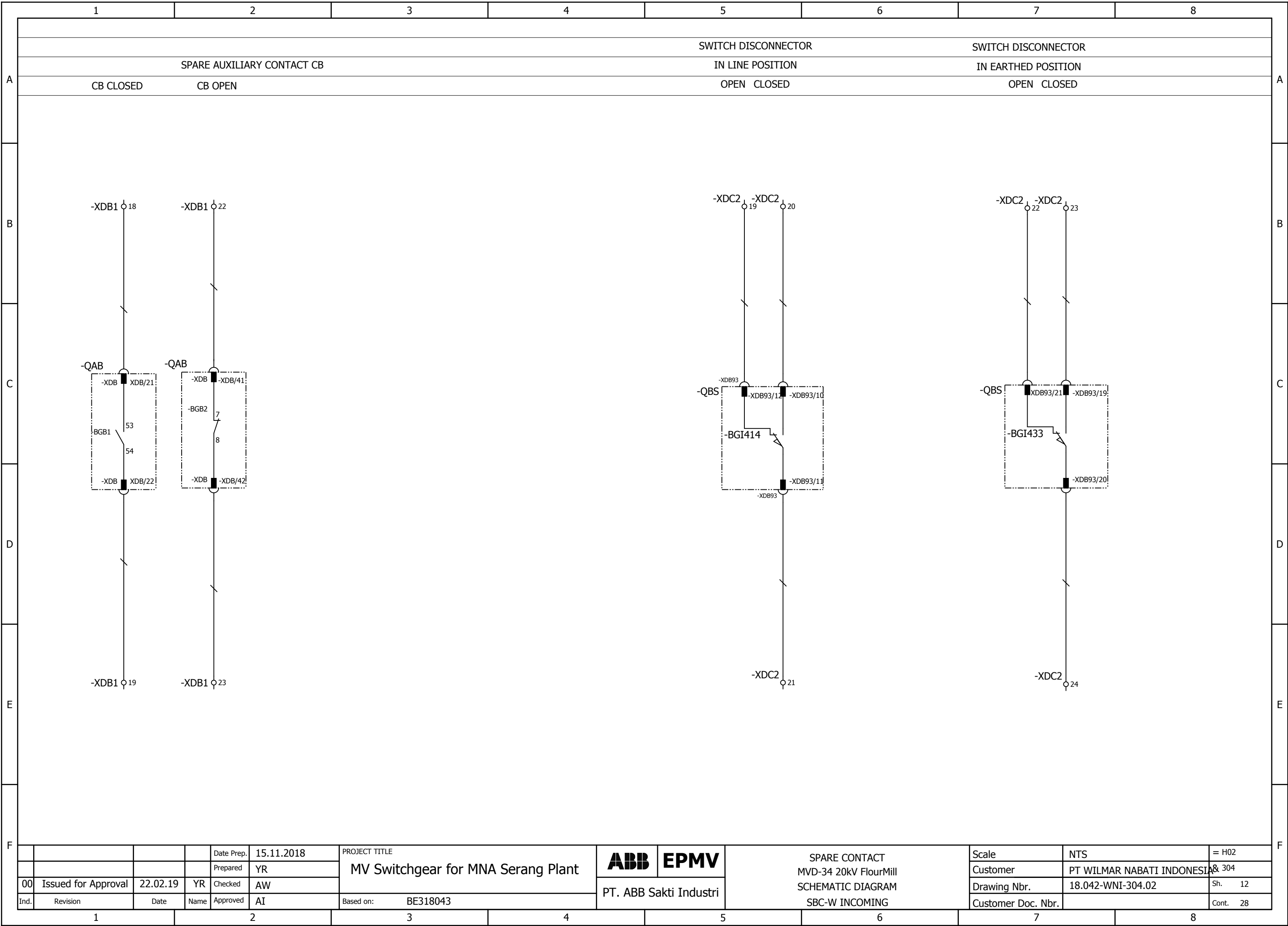
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|------|---------------------|----------|------|------------|------------|---------------------------------------------------------|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|----------|----------------------------|-------|
|      |                     |          |      | Date Prep. | 15.11.2018 | PROJECT TITLE<br><br>MV Switchgear for MNA Serang Plant |  |  | AUXILIARY MOTOR & RELAY<br>MVD-34 20kV FlourMill<br>SCHEMATIC DIAGRAM<br>SBC-W INCOMING | Scale    | NTS                        | = H02 |
|      |                     |          |      | Prepared   | YR         |                                                         |                                                                                       |                                                                                       |                                                                                         | Customer | PT WILMAR NABATI INDONESIA | & 304 |
| 00   | Issued for Approval | 22.02.19 | YR   | Checked    | AW         |                                                         | Drawing Nbr.                                                                          | 18.042-WNI-304.02                                                                     |                                                                                         | Sh. 10   |                            |       |
| Ind. | Revision            | Date     | Name | Approved   | AI         | Based on:                                               | BE318043                                                                              | Customer Doc. Nbr.                                                                    |                                                                                         |          | Cont. 28                   |       |
|      |                     |          |      |            |            |                                                         | PT. ABB Sakti Industri                                                                |                                                                                       |                                                                                         |          |                            |       |

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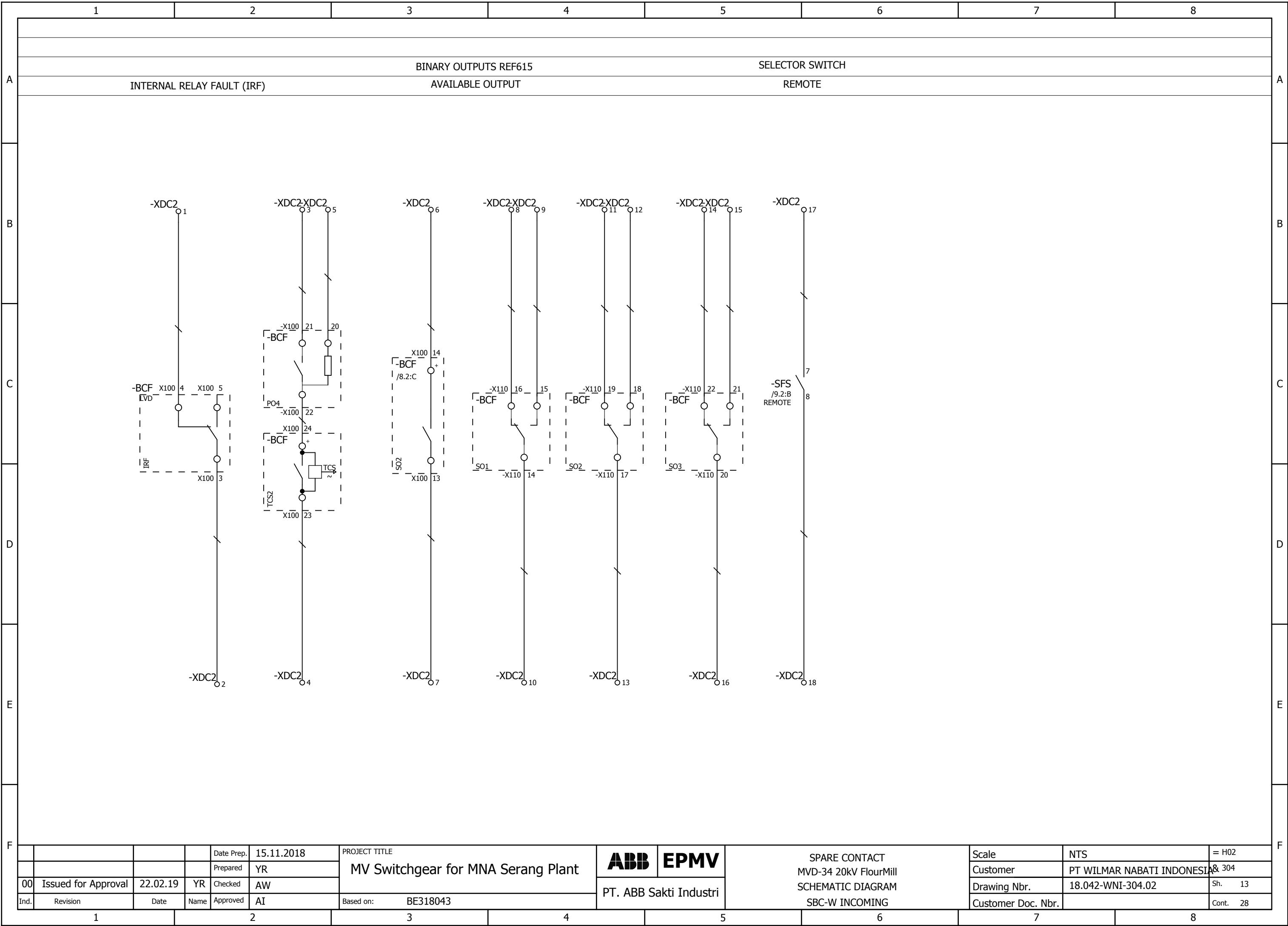


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|      |                     |          |      | Date Prep. | 15.11.2018 | PROJECT TITLE<br><br>MV Switchgear for MNA Serang Plant | ABB                    | EPMV         | DIGITAL INPUT/OUTPUT RELAY<br>MVD-34 20kV FlourMill<br>SCHEMATIC DIAGRAM<br>SBC-W INCOMING | Scale             | NTS                        | = H02 |
|      |                     |          |      | Prepared   | YR         |                                                         |                        |              |                                                                                            | Customer          | PT WILMAR NABATI INDONESIA | & 304 |
| 00   | Issued for Approval | 22.02.19 | YR   | Checked    | AW         |                                                         | PT. ABB Sakti Industri | Drawing Nbr. |                                                                                            | 18.042-WNI-304.02 | Sh.                        | 11    |
| Ind. | Revision            | Date     | Name | Approved   | AI         |                                                         |                        | Based on:    |                                                                                            | BE318043          | Customer Doc. Nbr.         |       |

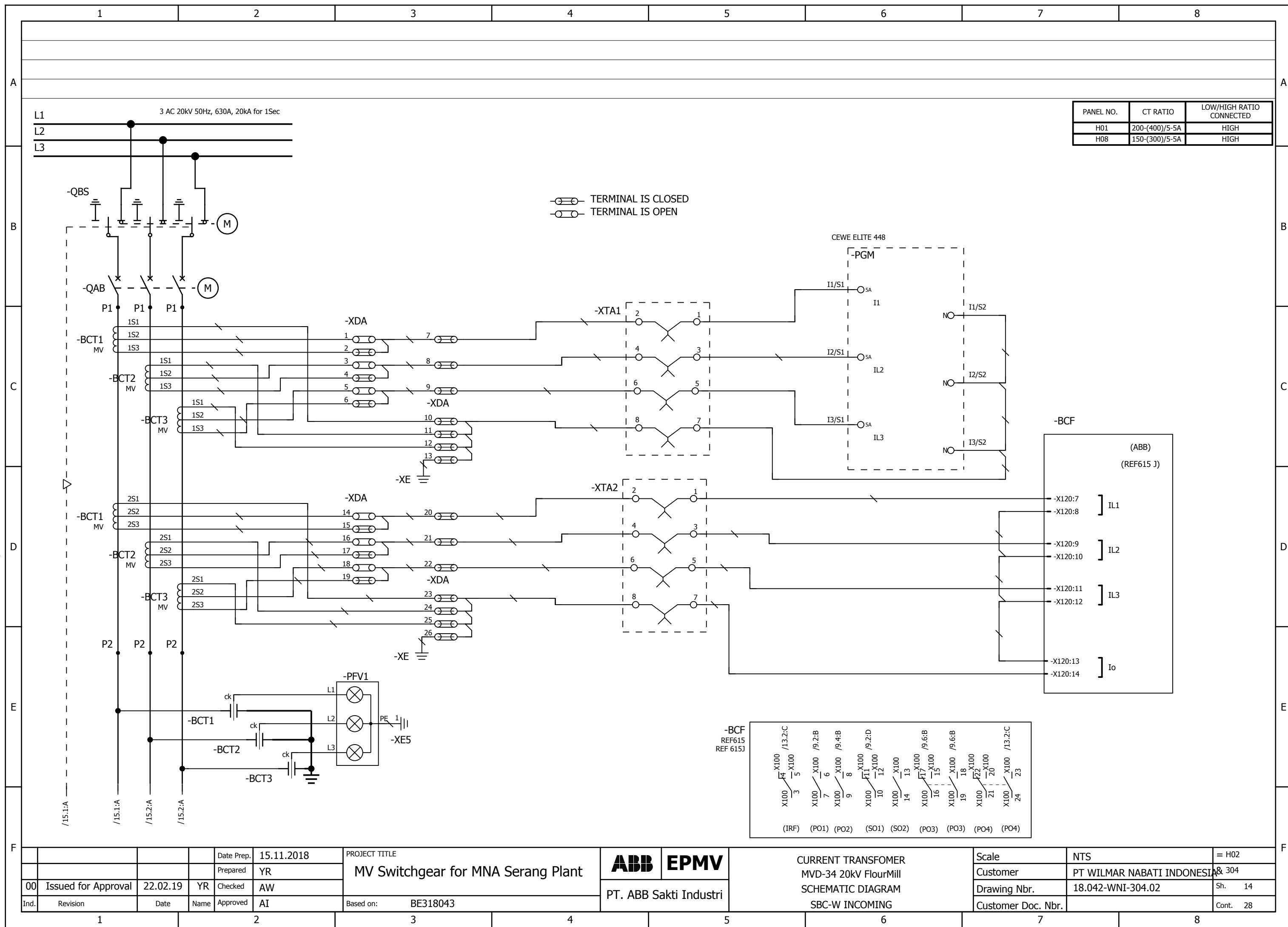
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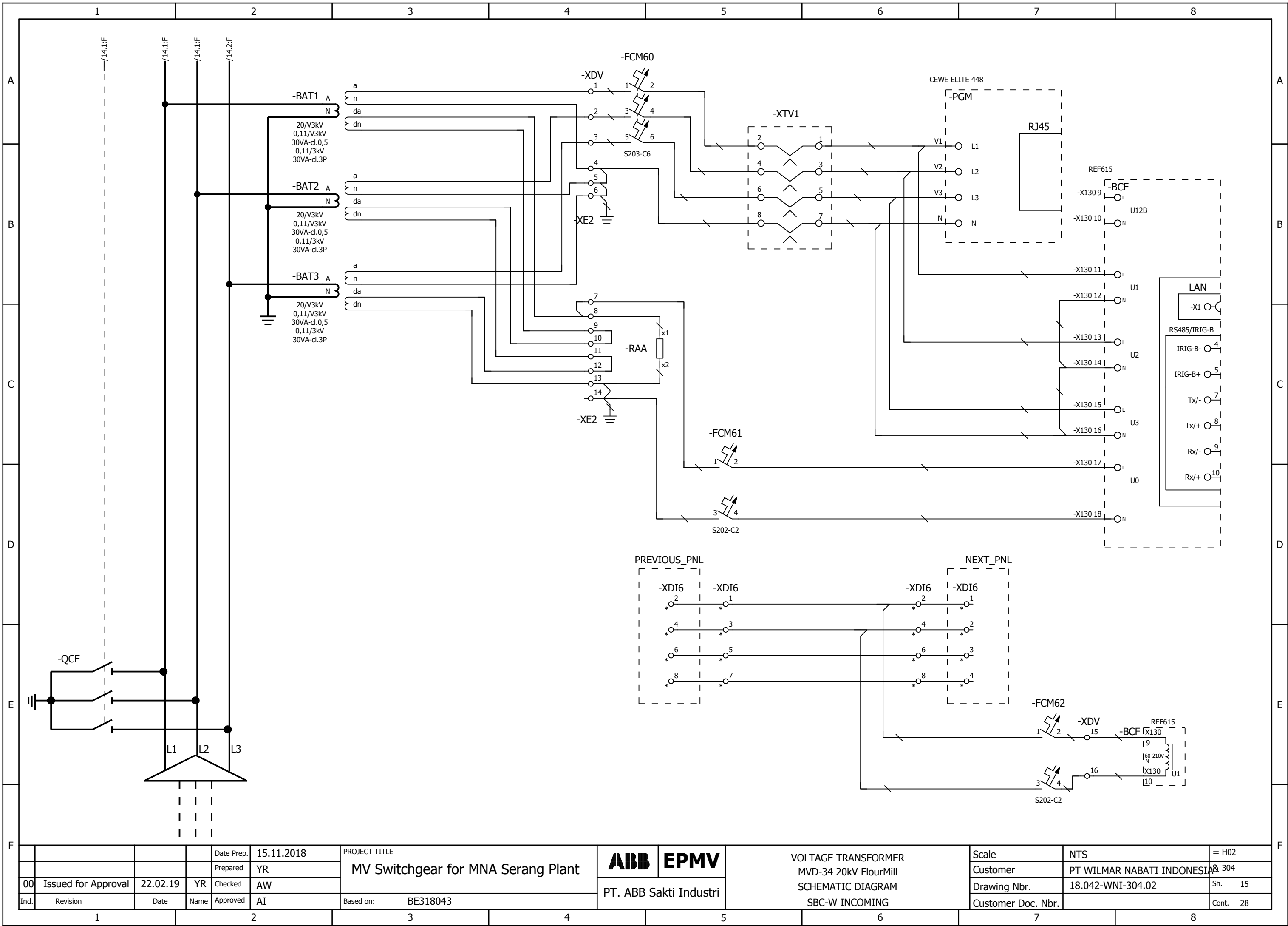


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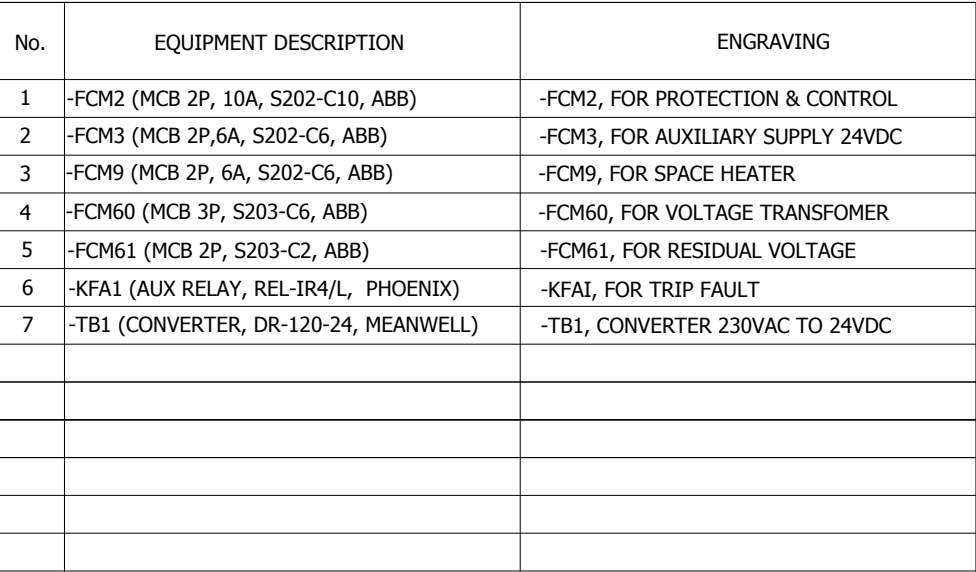


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| No. | EQUIPMENT DESCRIPTION                            | ENGRAVING                 |
|-----|--------------------------------------------------|---------------------------|
| 1   | -SFS (LOCAL-OFF-REMOTE SELECTOR SWITCH, K&N)     | -SFS, LOCAL-OFF-REMOTE    |
| 2   | -SFC (PUSH BUTTON CB CLOSE, GREEN, ABB)          | -SFC, CB CLOSE            |
| 3   | -SFO (PUSH BUTTON CB OPEN, RED, ABB)             | -SFO, CB OPEN             |
| 4   | -PFY1 (LAMP INDICATION FOR CB TRIP, YELLOW, ABB) | -PFY1, CB TRIP            |
| 5   | -PFR1 (LAMP INDICATION FOR CB CLOSE, RED, ABB)   | -PFR1, CB CLOSE           |
| 6   | -PFG1 (LAMP INDICATION FOR CB OPEN, GREEN, ABB)  | -PFG1, CB OPEN            |
| 7   | -PGJ (DIGITAL MEASUREMENT, CEWE 448)             | -PGJ, DIGITAL MEASUREMENT |
| 8   | -BCF (FEEDER PROTECTION RELAY, REF615J, ABB)     | -BCF, RELAY PROTECTION    |
| 9   | -XTA1, XTA2 (TEST BLOCK , YONGSUNG)              | -XTA, CURRENT TEST BLOCK  |
| 10  | -XTV1 (TEST BLOCK , YONGSUNG)                    | -XTV, VOLTAGE TEST BLOCK  |
|     |                                                  |                           |
|     |                                                  |                           |
|     |                                                  |                           |





|      |                     |          |      |            |            |                                                         |              |                    |                                                                               |                        |          |                            |
|------|---------------------|----------|------|------------|------------|---------------------------------------------------------|--------------|--------------------|-------------------------------------------------------------------------------|------------------------|----------|----------------------------|
|      |                     |          |      | Date Prep. | 15.11.2018 | PROJECT TITLE<br><br>MV Switchgear for MNA Serang Plant | ABB          | EPMV               | LV BOX LAYOUT<br>MVD-34 20kV FlourMill<br>SCHEMATIC DIAGRAM<br>SBC-W INCOMING | Scale                  | NTS      | = H02                      |
|      |                     |          |      | Prepared   | YR         |                                                         |              |                    |                                                                               | PT. ABB Sakti Industri | Customer | PT WILMAR NABATI INDONESIA |
| 00   | Issued for Approval | 22.02.19 | YR   | Checked    | AW         |                                                         | Drawing Nbr. | 18.042-WNI-304.02  |                                                                               |                        | Sh.      | 17                         |
| Ind. | Revision            | Date     | Name | Approved   | AI         | Based on:                                               | BE318043     | Customer Doc. Nbr. |                                                                               |                        | Cont.    | 28                         |

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|-----------------------|--|-----------------------------------------------------------------------------------------------------------------|--|--|--|------------------|--|----------------------------|--|----------|--|-----------------|--|---------------------|--|--|--|
| Summarized parts list |  |                                                                                                                 |  |  |  |                  |  |                            |  |          |  |                 |  | IDABB_F02_005-IDABB |  |  |  |
|                       |  |                                                                                                                 |  |  |  |                  |  |                            |  |          |  |                 |  | Carry over: 0.00    |  |  |  |
| TAG ID                |  | DESCRIPTION                                                                                                     |  |  |  | MANUFACTUR/BRAND |  | PART NUMBER                |  | QUANTITY |  | REMARKS         |  |                     |  |  |  |
| -BCF                  |  | Feeder Management Relay<br>HBFJAEAGABC1ANA11G                                                                   |  |  |  | ABB              |  | REF615J HBFJAEAGABC1ANA11G |  | 2        |  | /8.2:C;/15.7:B  |  |                     |  |  |  |
| -EB                   |  | HEATER<br>230VAC, 50W                                                                                           |  |  |  | SANYANG          |  | CDXN010033P0009            |  | 1        |  | /7.6:E          |  |                     |  |  |  |
| -FCM2                 |  | MCB S202-C10<br>MCB. 2P, 10A                                                                                    |  |  |  | ABB              |  | 2CDS252001R0104            |  | 1        |  | /7.2:C          |  |                     |  |  |  |
| -FCM3;-FCM9           |  | MCB S202-C6<br>MCB. 2P, 6A                                                                                      |  |  |  | ABB              |  | 2CDS252001R0064            |  | 2        |  | /7.5:C;/7.6:C   |  |                     |  |  |  |
| -FCM60                |  | MCB S203-C6<br>MCB. 3P, 6A                                                                                      |  |  |  | ABB              |  | 2CDS253001R0064            |  | 1        |  | /15.4:A         |  |                     |  |  |  |
| -FCM61;-FCM62         |  | MCB S202-C2<br>MCB. 2P, 2A                                                                                      |  |  |  | ABB              |  | 2CDS252001R0024            |  | 2        |  | /15.5:D;/15.7:E |  |                     |  |  |  |
| -KFA1                 |  | Auxiliary Relay<br>230AC 4COS                                                                                   |  |  |  | PHOENIX          |  | 1YRA3DL266230001           |  | 1        |  | /9.7:E          |  |                     |  |  |  |
| -PFG1                 |  | PILOT LIGHT GREEN #CL-523G#230VAC<br>CL-523G                                                                    |  |  |  | ABB              |  | 1SFA619402R5232            |  | 1        |  | /8.5:E          |  |                     |  |  |  |
| -PFR1                 |  | PILOT LIGHT RED #CL-523R#230VAC<br>CL-523R                                                                      |  |  |  | ABB              |  | 1SFA619402R5231            |  | 1        |  | /8.4:E          |  |                     |  |  |  |
| -PFY1                 |  | PILOT LIGHT YELLOW #CL-523Y#230VAC<br>CL-523Y                                                                   |  |  |  | ABB              |  | 1SFA619402R5233            |  | 1        |  | /8.6:E          |  |                     |  |  |  |
| -PGM1                 |  | POWER METER CEWE 448<br>Class : 0.5s; CT Input : 1A; VT Input : 57.7-240V (L-N); Aux. Voltage: 80-300VAC/DC     |  |  |  | CEWE             |  | 1YRA4EA0350001             |  | 1        |  | /8.3:C          |  |                     |  |  |  |
| -SFC                  |  | PUSH BUTTON, YW1B-M1E10-G<br>GREEN COLOUR                                                                       |  |  |  | IDEC             |  | 1YRA3DA14700001            |  | 1        |  | /9.2:B          |  |                     |  |  |  |
| -SFO                  |  | PUSH BUTTON, YW1B-M1E10-R<br>RED COLOUR                                                                         |  |  |  | IDEC             |  | 1YRA3DA14700002            |  | 1        |  | /9.4:B          |  |                     |  |  |  |
| -SFS                  |  | LOCAL/REMOTE Selector Switch<br>Type : S0-CA10-A723-600-E; 4NC (for LOCAL Position) & 4NO (for REMOTE Position) |  |  |  | K&N              |  | 1YRA3DA17940011            |  | 1        |  | /9.2:B          |  |                     |  |  |  |
| -TB1                  |  | Voltage Input : 85-264 VAC/120-370V DC<br>Voltage Output : 24 VDC, 5A                                           |  |  |  | MEANWELL         |  | DR-120-24                  |  | 1        |  | /7.3:D          |  |                     |  |  |  |
| -XTA1;-XTA2           |  | Current Test Block<br>Rated Current : 10A, 250V; Mounting : Flush Mounting                                      |  |  |  | YONGSUNG         |  | YSCTT-04C                  |  | 2        |  | /14.4:B;/14.4:D |  |                     |  |  |  |
| -XTV1                 |  | Voltage Test Block<br>Rated Current : 10A, 250V; Mounting : Flush Mounting                                      |  |  |  | YONGSUNG         |  | YSPTT-04C                  |  | 1        |  | /15.5:A         |  |                     |  |  |  |
|                       |  |                                                                                                                 |  |  |  |                  |  |                            |  |          |  |                 |  |                     |  |  |  |
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